

Bi-Co Food System Report

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Introduction

In the field of Environmental Studies, food systems are a critical topic because where, how, and when we get our food is deeply interrelated with the environment. While many are unaware of this, there are several major steps that food has to go through before we are able to consume it. Each step has an impact on the environment. According to the United Nations,

Food needs to be grown and processed, transported, distributed, prepared, consumed, and sometimes disposed of. Each of these steps creates greenhouse gases that trap the sun's heat and contribute to climate change. About a third of all human-caused greenhouse gas emissions is linked to food (2025).

In other words, what we eat on campus can have a global effect, as each part of the broader food system negatively affects the environment. As the impact of climate change continues to increase, it is crucial that we understand the role that our eating habits play in this crisis.

Furthermore, Environmental Studies is connected to food justice and food scarcity. Food waste, climate change, and social conflicts are all contributing to food becoming more scarce. Currently, one in eleven people globally face hunger every day (United Nations, 2025). In 2018, the EPA estimated that in the United States, food contributed to landfills and combustion facilities more than any other material (Adams, 2024). Clearly, food waste plays a big role in food scarcity. Moreover, climate change is impacting our agricultural systems as natural disasters increase in frequency and strength. Since our agricultural systems are fragile and dependent on a healthy environment to produce food effectively, climate change has devastatingly led many agricultural crops to die (Adams, 2024). Additionally, with pandemics and international conflicts, many countries have had to shift their priorities away from food production. These

types of social issues also stretch food availability, often resulting in even more food shortages (Adams, 2024).

Finally, it is well known that many low-income urban communities in the United States lack access to healthy and affordable food. Many times, the only food sources in these areas are convenience stores that offer unhealthy snacks and barely any fresh produce (Zaidi 2024).

Besides, even if there were fresh produce available, it would be sold at a higher price. According to a 2005 study by Cheryl Brown and Mark Sperow, an all-organic diet costs consumers a 49% price premium. It is safe to assume that this cost has only increased due to inflation. Increasing access to healthy affordable food is a primary issue related to food justice. Grassroot organizations are working to address how geography and socio-economic factors play a large role in how much and what types of food people have access to.

In our introduction, we have provided information on some of the main concerns experts are working to address within the fields of Food Studies and Environmental Studies to highlight how food is involved in many aspects of our lives.

Our project

After spending four years at Bryn Mawr College and Haverford College, we find ourselves still wondering how food on campus operates. No one seems to know the full picture of each campus's food system. For our senior Environmental Studies capstone, we decided to do a thorough investigation of places within the Bi-Co where we can prepare our own food, access ready-to-eat meals, purchase snacks, and supply ourselves with the materials needed to cook. With this project, we hope to piece together information to share with other students, staff, faculty, and community members within the Bi-Co.

Among the four of us working on this project, we found that our relationship with food on campus varies greatly. Some of us live in an apartment where we cook our own food, some are on a residential meal plan, and others are low-income and rely on organizations like The Nest as a supplement to the meal plan. Sharing these differences with one another, we realized how important it is to know how food systems in the Bi-Co are connected and how they operate simultaneously. This led us to focus our project on mapping the stakeholders, locations and availability of different food sources on Haverford and Bryn Mawr's campuses.

The skills we have learned throughout our time in the Environmental Studies major prepared us to create this project. We have learned that looking at food related issues requires complex and interdisciplinary skills, research, and solutions. Therefore, we employed three different mapping styles to represent these systems in different ways; the mapping styles include a GIS, systems, and participatory map. To create these maps, we had to develop a research plan.

Our plan ended up including many steps. First, we examined the past of the Bi-Co food system and collected information online and at Special Collections about how food on campus has looked. Then, we conducted interviews with food stakeholders on the two campuses to get firsthand information about the many places that involve food in some way. That information is represented through a “story map” on ArcGIS, allowing viewers to scroll through and visually explore the physical relationship between the food locations on Haverford and Bryn Mawr’s campuses. Next, those interviews were used to create a “systems map” inspired by Donella Meadows, allowing viewers to see a less place-based perspective on relationships between stakeholders on and off campus. Finally, we also employed “participatory mapping” as a way to crowd-source how Haverford students interact with food sources on and off campus.

These three mapping techniques are used in conjunction with each other to accomplish our goals of 1. uncovering potentially hidden or unknown information about the Bi-Co food system, 2. analyzing the state of the system, and 3. providing recommendations about how to improve its efficiency, effectiveness, and sustainability.

History: The development of these institutions' food systems

I. Haverford

Haverford College was established in 1833 by the Society of Friends. There have always been places to get food on campus, but they have changed and evolved over time. In this section we give an overview of some of the places one can get food on campus today, and the history of those places. Until 1882, the dining hall was located in the basement of Founders Hall. It was eventually moved onto the first floor of the building. However, it was moved again in 1907 to the Great Hall until 1969 when the current Dining Center was built (Ross et al.).

Another building that provides food for students is Whitehead Campus Center, which opened in 1993. Inside the building, there are two kitchens, one of which is specifically Kosher, both located on the second floor. Whitehead is also the home of the Coop. Originally, this dining facility was known as the Coffee Shop and served pizzas and gourmet coffee (HCB-001-015). Now in 2026, the Coop no longer serves fresh coffee, but it does serve a variety of made to order items like breakfast fare, sandwiches, and more. Furthermore, Whitehead is also home to The Nest, a food pantry combating food insecurity amongst low-income students. Initially, the Nest was the game room but was repurposed in 2020.

In addition, there is a cafe known as the Library Cafe, located on the ground floor of Lutnick Library. The library has been remodeled and constructed many times. In the most recent renovation in 2020 (Dean Rogers), the cafe was added. The Library Cafe provides coffee, lattes, cappuccinos, teas, fresh fruit, and grab and go breakfast and snack items.

Lastly, there are several kitchens on campus in addition to the two in Whitehead. Located in the Visual Communications Arts and Music (VCAM) is the VCAM kitchen. VCAM was once the campus's gymnasium, but it was renovated in 2017 with the goal of creating a space for students to be creative. During this renovation, the VCAM Kitchen, which may be used for personal reasons or class, was added to the space. Another kitchen is located in the Multicultural Center room in Stokes. We were unable to find official information about this kitchen, but it is also available for students to reserve and use.

Like many other institutions, Haverford also has several vending machines across campus. Besides the vending machine in the Dining Center basement which serves smoothies, all the vending machines on campus offer snacks and sodas.

The Dining Center, the Coop, and the Library Cafe remain the primary sources of food for students on campus. With the additions of spaces like the VCAM Kitchen and The Nest, students have more options and authority to prepare their own food.

Brief History of Student Engagement at Haverford

Food access and quality have long been a topic of interest for students at Haverford College. In 2008, a student club called Food Fight emerged to bring local and organic food into the Dining Center (HCQ.003.112). Later in 2013, another food-centered club formed, known as the Food Justice Coalition, which wanted to integrate the Real Food Challenge, a national

movement for college campuses to spend 20% of their budgets on local, humane, fair trade, and ecologically sound food by 2020. They worked with the Dining Center to order such food based on previous spending habits (HCQ.003.081, HCQ.003.130). After speaking with Tom Mitchell about updates on the Real Food Challenge, I found that the goals asked by the initiative were met well before the 2020 goal.

Campus clubs and organizations continue to grow at Haverford and adapt to the ever-changing food systems by responding to unique dietary needs, food insecurity and sustainability concerns. Present day initiatives such as The Nest and EHaus are fully student-led and work to increase food justice on campus through access to food. We believe that a fundamental step to changing the food system is understanding it. We hope that the information that we collect in this project can give students the information they need to continue to advocate for and make change.

ii. Bryn Mawr

Bryn Mawr College was established in 1885 as an all women's college to offer women the chance to have a rigorous experience in higher education ("College History & Legacies," n.d.). In this section, we give an overview of some of the places where one can get food or that produces food on campus today, and the history of those places. One of the most important things to know about the early days of the college's food systems is that every dorm had its own dining hall. Merion Hall was the first dorm to open in 1885 and thus, had the first dining hall on campus ("Merion," n.d.). The two dining halls on campus today, Erdman and New Dorm Dining Halls, opened in 1965 and 2015 respectively ("Erdman," n.d.; "New Dorm," n.d.). The college

shifted from having dining halls in each dorm to two centralized locations because of staffing shortages.

The college's community garden opened in 2010 and was founded by the student-run Green Gardens Committee in collaboration with the Facilities department (Weber, 2023). Everything that is grown in the garden is free to students, who can go to the garden whenever they want to and pick things to eat. The produce is not distributed to other places on campus right now, but there is hope that in the future produce grown in the garden can go to the dining halls and UnCommon Grounds, a cafe on campus. The garden is currently undergoing renovations thanks to two student garden coordinators, who are being funded by the Sustainability Office's Seed Grant Program. New beds have been put in, and the garden reopened to the community on Earth Day 2026.

There is only one kitchen available to students on campus, which is the Well Community Kitchen that opened in 2022 (Gray, 2022). Attached to the Well Community Kitchen is a Kosher kitchen and a Halal kitchen, both of which can only be entered with keys obtained after students complete the necessary training on how to respect the religious standards of each respective kitchen. Anyone in the Bryn Mawr College community can book time to use the Well Community Kitchen, though, to prepare and cook food. Staff members frequently use the kitchen during working hours. The Administrative Assistant of Conferences and Events, Debra Chadwick, shared with us that about ten students book the kitchen each week at night, mostly to cook individual meals. Occasionally, people use the kitchen for a group gathering like a birthday party or a class event. The kitchen is supplied with very few things, so individuals need to bring their own supplies and food.

As with Haverford, vending machines are present throughout Bryn Mawr College's campus. Bryn Mawr has more vending machines than Haverford, though, as there are vending machines in most academic buildings at Bryn Mawr but not at Haverford. Dining services manages them and provides a list of where each vending machine is located on their website ("Vending," n.d.). In 2024, Park Science Center got a Farmer's Fridge vending machine, which serves healthy salads, wraps, bowls, and snacks (Clow, 2024). The other vending machines on campus serve packaged snacks and drinks.

While Erdman Dining Hall and New Dorm Dining Hall remain the most common places people get food, there are also two cafes on campus. Unfortunately, the history of these cafes was not accessible to us during the time we completed this project.

While it is helpful to know about these campuses' pasts, the rest of our paper will focus on the current role of food related locations on the campuses of both Bryn Mawr and Haverford.

Methods: How did we gather information on the current Bi-Co food system?

Data Collection: Interviews

We determined that the most effective way to find out information about the Bi-Co food system was to interview stakeholders associated with all of the locations we included in our project. As students, we have heard from many of our peers that it is hard to get in touch with staff members, so it was important to us that we worked to reduce this barrier by trying to establish relationships. We also wanted to talk directly to stakeholders because those who work at these locations can provide firsthand information, which would give us insights that others

likely do not know. The information collected from the interviews informed our ArcGIS map and our systems map.

1. Determine relevant locations to research:

In order to conduct interviews, we first had to figure out which locations we felt should be included in our project. We used our collective knowledge of both of the campuses to figure out which locations either prepare, grow, provide, sell, serve and/or dispose of food. It was important to us that we included locations that collectively fell under all of these categories because food systems involve every step of the process that food is involved in, not just where food is served.

The locations we decided on are as noted below:

Haverford College

- The Dining Center
- Library Cafe
- The Coop
- The Nest
- Haverfarm
- VCAM Kitchen
- Ehaus
- HCA Garden
- CER Composter
- Arboretum
- Vending Machines

Bryn Mawr College

- Erdman Dining Hall
- New Dorm Dining Hall
- UnCommon Grounds

- Lusty Cup
- The Well Kitchen
- Community Garden
- Tea Pantries
- Vending Machines
- ECC Kitchen
- Batten House
- Wyndham

2. *Data collection through interviews*

The next step in our process was to figure out who we should interview. For each of the locations, we found a staff member or student to reach out to with an interview request. For the places that are staff run, we first reached out to staff members who were listed on each of the college's websites as involved with these places before reaching out to students. If the staff member did not answer, we followed up. If they still did not answer, we tried to reach out to other staff members, when applicable, and as a last resort, we would reach out to students who worked at these places. For places that are student run, we reached out to students first. Overall, who we reached out to first depended on who was the person we expected to have the most firsthand experience with the location.

We then compiled a list of questions to ask each of the people we requested an interview with. Some of the questions we asked included:

- How do you see your organization fitting into the food web of Haverford?
- What food do you create or use?
- What facilities or resources for food do you provide?
- What services do you provide?
- Who is your audience, who do you serve?
- What is challenging?

- What do you feel could be improved?
- What directions are you growing in?

We also asked questions unique to each location to get a more detailed description of what each part of the food system looks like. For example, these are some questions we asked for the vending machines on both campuses:

- Why are they placed in that precise location?
- Price range of items?
- What are they selling? Is it healthy?
- Where does the food come from?
- Who decides what is being sold in them?
- How do they make all these decisions?

Since we split up who conducted each interview, notes were taken in a collective document, so we all could see the information that was collected. We also included notes below the interview notes from observational research (what we noticed when we were at these places) and photo collections.

This information was used to create the ArcGIS map and the systems map of the Bi-Co food system. Both qualitative and quantitative data appears in the ArcGIS map, while only qualitative data appears in the systems map.

GIS Map

With the lack of transparency about food spaces on Bryn Mawr and Haverford's campuses, we thought it would be useful to make a ArcGIS map that shows the places involved in these colleges' food systems. An ArcGIS map is useful in particular because it attaches information to a spatial coordinate, which helps people understand the areas on campus we are talking about. Even if someone does not know the name of the location we are talking about,

they can see where the space is physically located on the map. Using either context clues about the location of the space or by actually going there to check it out, Bryn Mawr and Haverford community members can easily learn about food spaces on these campuses through our map.

Making the map itself included several steps. First, building outlines for both campuses were downloaded from Montgomery County and Delaware County GIS online data hubs because those are the counties the colleges are located in. These data files were uploaded into ArcGIS. Then, the building outlines of the locations we included in this project were selected to remain visible on the map. Each of these buildings was then downloaded, so they could be added to ArcGIS Online, which is a platform that makes maps accessible to everyone. We decided to use the story map feature on ArcGIS Online to display our map because it provides different options for how people can engage with the content in the map. Our story map includes the option to click on a location and see the attached information or to scroll through every location on a map tour. We decided to include both of these options because it gives people viewing our map the ability to quickly access information for a specific building by clicking on it or to get a broad picture of the whole food system by scrolling through the map tour. Overall, our map aims to share the information we collected in an accessible, spatially relevant manner.

Systems Mapping

Earlier in the semester, our class read Donella Meadows' *Thinking in Systems*. In the introduction, Meadows states, "Once we see the relationship between structure and behavior, we can begin to see how systems work, what makes them produce poor results, and how to shift them into better behavior patterns" (Meadows 1). It is with this objective that we brought systems mapping into our project. Not concerned with the physical location of things, systems

mapping allows us to track what food and food-related resources are "flowing" in and "flowing" out of our Bi-Co system in a way that other forms of mapping can't. While we aren't claiming that the Bi-Co system is producing "poor results," we are excited to potentially reveal "how to shift them into better behavior patterns" (Meadows 1).

Gathering information for systems mapping started with some of the interview questions we asked earlier in the process such as "what facilities or resources for food do you provide?" and "what's coming in?/what's coming out?". After conducting the interviews, we cleaned up the information into sections for the story map and included a section called "potentially relevant information for systems mapping." We then took these blurbs for every location we studied and started drafting a systems map on Canva. Red rectangles represent all of the food locations we researched; green rectangles represent off campus vendors; yellow rectangles represent funding sources; and blue circles represent beneficiaries of the food system. Arrows represent resources constantly "flowing in" and "flowing out" of different parts of the system. Although the maps do not perfectly represent all of the many complicated relationships between parts of the complex Bi-Co food system, it is beneficial to see patterns emerge.

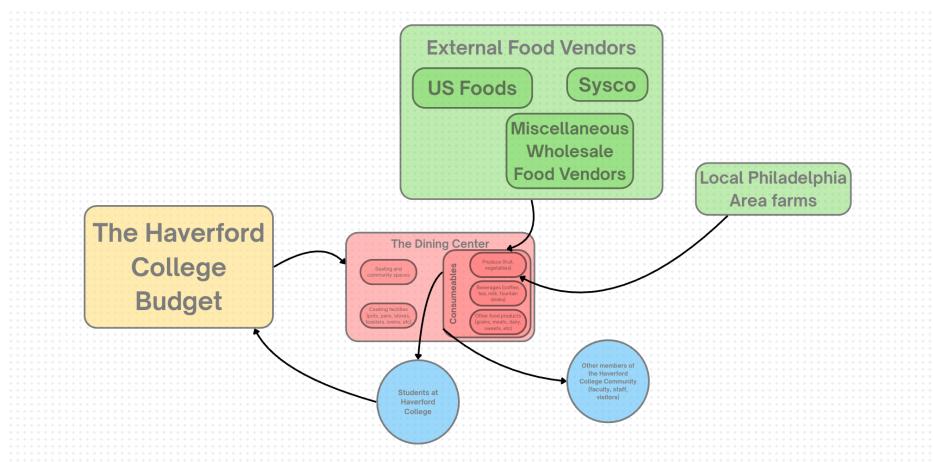


Figure 1: Simplified Systems Map of the Haverford Dining Center

Participatory Mapping

While our GIS map overviews important food locations and student organizations on each of these campuses, and the systems map tracks how resources flow between locations, we still did not know how students interacted with these spaces or food oriented spaces *off* campus. Considering off campus in food relations is important because Haverford College has 21 apartment buildings as on campus housing that all have full kitchens, and approximately 500 students live in the apartments. Many students who live in the apartments are on partial meal plans, meaning that they grocery shop and cook for themselves in addition to eating at the dining halls.

Given the time constraints of the project, our lack of experience creating large surveys, and our interest in mapping how students interacted with the different sites, we decided to use participatory mapping, which is a method “in which community members and researchers or practitioners apply their collective knowledge to produce maps that represent areas of importance” (GEMS). Participatory mapping also highlights community members’ “interests, expectations, perceptions, and experiences with reference to specific geographic landmarks” (GEMS). Through participatory mapping, we were able to gather data on where students get their food on and off campus. It is also interesting to note, as we will talk about more later on, the participatory map did not have to be precise in order to still provide interesting and relevant data.



Figure 2: Students Participating in Participatory Mapping

To learn about where students get their food off campus, we selected two 30 minute blocks when we expected there would be a high volume of students walking to and from the Haverford Apartments. To ensure that we were mostly sampling students who live in the apartments, we set up our survey station in a bulletin board hut by the entrance to the apartment complex. We recruited students by asking them “where do you get your food?!” as they walked by. If the location was within the geographical confines of the map we printed out, we placed an ‘X’ on the location, if it was not, we wrote the location down and added a ‘check’. Most people that passed by answered, and some even came over to mark the map themselves. However, students on the phone or groups of athletic men generally did not respond. We used an online counter to track the number of people who responded because respondents frequently marked multiple places on the map.

To learn about where students get their food on campus, we used a similar method. We chose a 30 minute block, this time in Haverford College’s Dining Center. We located ourselves here because it was topical, and we wanted to interview students who lived all over Haverford

campus. As students walked by, we asked them where they got their food and followed the steps noted above.

Results: What we found

Our Website

Check out [our website](#) to see all of our maps with the information we collected!

GIS Map

As displayed by our ArcGIS map, the Bi-Co food system spread across the entirety of their campuses. There are a variety of food spaces, which include ones that are part of dining services, gardens, student-run kitchens, tea pantries in the dorms, etc. In our map, one can clearly see that at both Bryn Mawr and Haverford, the dining halls are on one side of campus and the campus cafes—UnCommon, Lusty Cup, The Library Cafe, and The Coop—are located on a different part of their respective campuses. At Haverford, meal swipes can be used at The Coop and The Library Cafe, but at Bryn Mawr the meal plan does not pay for anything at UnCommon or Lusty Cup. This is a particularly interesting finding to address because if students live on the part of these campuses that are closer to the cafes, they are more likely to get their food from there than from the dining hall. Seeing this mapped out only further brings up concerns about the accessibility of the cafes at Bryn Mawr. Students who live close to UnCommon might not be able to afford to buy their food from there each day and might have to go out of their way to get to the dining halls, which are both located on a separate part of campus than where the cafes are.

Another aspect of these food systems that is highlighted by our map is the evolving presence of student kitchens on both campuses. Within the past ten years, both campuses have added a community kitchen: VCAM and the Well Kitchen. Students living in dorms use these kitchens to cook homemade meals for themselves and/or professors/staff can rent out these kitchens to hold classes and community events. Our research revealed that the need for community kitchens came from the fact that cooking appliances in dorms have become more and more limited over time. At Bryn Mawr, there used to be full dining halls in each dorm, then there was just a kitchen, then there was just an oven, and today there is just a bread toaster and microwave. Staff members shared that the reason behind these changes in the food system came from staffing shortages and concerns about fire safety. The current solution to the limited ability for students to cook in dorms is to add community kitchens to both campuses.

Furthermore, the information contained in our ArcGIS map highlights how the dining halls have had to adapt to serving a variety of foods to meet all students' needs. This is one of the main things on the minds of those who manage the Dining Halls. Richard Clow, Associate Director for Administration of Dining Services at Bryn Mawr, shared with us that they now work with over 60 vendors in order to get a variety of foods that students desire. Richard also told us that they are working to reopen Rhoads Dining Hall in order to have an allergy free kitchen, where students with food allergies can better access the food they need. With the growing needs of students, infrastructure is one of the biggest challenges they are facing to be able to expand dining services. As our map shows, the dining halls at both campuses have evolved a lot over time, and they are continuing to change each day.

The information shared in the map also emphasizes the huge role students play in shaping these food systems. Almost every one of the locations highlighted mentions how decisions about

what food is prepared, grown, and/or served is based on the desires of students. If students have feedback, they adjust what they are preparing, growing, and/or selling to better fit students' needs. One channel for feedback is within Bryn Mawr College's dining halls, where there is a board for students to leave notes for the dining hall workers on what they liked and what should be improved; they call it "napkin notes". Dining Services at Haverford College offers an online survey, where students can share their feedback on the Dining Center. Interacting with our map will teach our community about how involved students truly are in the Bi-Co food system.

Our map shares the basics of how food decisions are made at each place we include, where the food comes from, and what types of food and people these places mainly serve. Students, faculty, staff, and community members who do not know about how Bryn Mawr and Haverford College's food systems function can use our map to get an understanding. Sources are included for every location, so especially interested individuals know who to contact if they wish to learn more.

Lastly, an interesting part of this work that we found out throughout this project is that there seems to be some hesitancy from staff members to talk about their role in the food system. Several of the people we reached out to interview, who were directly involved in food spaces on campus, referred us to talk to someone higher up like their bosses. This brings up questions about why people did not want to speak with us. Were they scared to have their words on record? Did they not have time? Are they told to refer any questions to their bosses? We do not know, but this reaffirmed for us that it is quite complicated to try to get full transparency about these food systems.

Systems Map

Guided by Donella Meadows, the process of developing the systems map revealed the complexity and interconnectedness of the Bi-Co food system. Through the process, some recurring patterns emerged that were significant: reciprocal relationships, resource sharing relationships, and resource loops.

Reciprocal relationships are where two parts of a system exchange different resources with each other and each benefit from the relationship. For example, students living in Ehaus bring groceries and labor into the Ehaus system and in return, they get a tasty meal! This is a pattern that occurs often in relationships where labor is required to create resources. Reciprocal relationships are important in a system because it creates strong and interdependent relationships between parts of a system, helping the system as a whole.

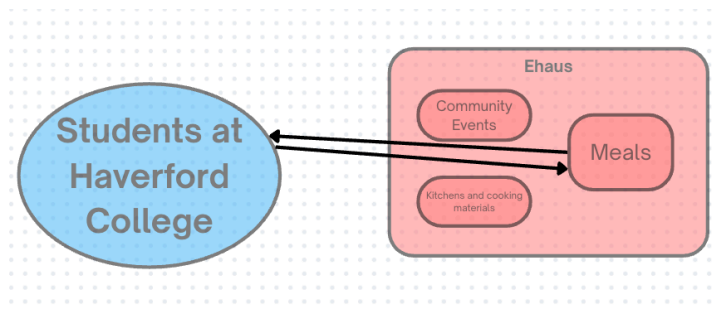


Figure 3: Reciprocal Relationship with Ehaus

Resource sharing relationships occur when two or more parts of a system are able to share resources with and among each other. For example, the Haverfarm shares its farming equipment and seeds with the HCA Garden Club. Although the Haverfarm doesn't get anything concrete in return, resource sharing is a sign of a healthy community. The Haverfarm may lose some of their resources, but they intangibly benefit from being a part of a strong system that includes the HCA Garden Club.

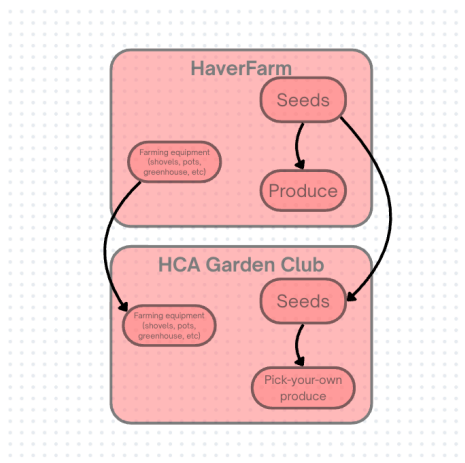


Figure 4: Resource Sharing Relationship with the Haverfarm and HCA Garden Club

Sometimes when systems mapping, loops emerge where many parts of a system exchange resources in a way that benefits each other and the system as a whole. For example, students pay tuition and that funding flows into the college budget which in turn, funds the operations of the Dining Center which in turn, provides food to the students. These types of relationships are very common in complex systems because each part of a system is not always able to exchange resources that would be useful to another part, so by getting more parts involved in the exchange, all parts of the system are able to receive useful resources. Money is quite often one of the resources involved in these relationships.

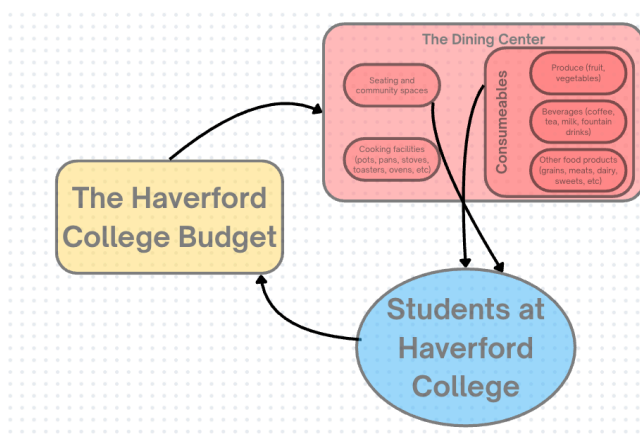


Figure 5: Resource Loop with the Dining Center, the Budget, and Students at Haverford College

Participatory Map

The participatory mapping section at the Haverford Dining Center was successful in the number of students, but not in the type of data that we collected. We talked to 124 students. The students mostly marked the Dining Center as the location from which they got food. A couple of students also included The Coop. When asked, people provided us with names of the places they eat off campus, but those restaurants were places they dined only occasionally, instead of every day like the Dining Center. Interestingly, while students who live in the apartments mentioned that they eat at the dining center, none of the students we talked to in the Dining Center mentioned grocery shopping or eating at the apartments as sites where they interact with food.

From the participatory mapping we did by the apartments, we got 84 student responses, which is about 16% of students who live in the apartments and 5% of total Haverford students. Because the 'X's were hard to count in large numbers, it is unclear precisely how many tallies each location received. However, Trader Joe's clearly had more tallies than any other location. Interestingly, almost every grocery store in the area was represented at least once. Although not reflected in our data recording method, we learned from conversations that some people primarily ordered groceries from a location instead of physically going. It also seemed that proximity played a large role in how and where people decided to shop. For example, one individual went to Walmart as part of their weekly commute home, while other people who wanted Walmart products generally opted to order them. Similarly, HMart and Weee, which provide a greater range of Asian grocery goods than the supermarkets in the area (but similar products across the two stores), had an equal number of patrons. However, HMart is generally an in-person experience while Weee is exclusively an online store which may be the deciding factor in how people choose which to go to.

We found this pattern also holds true for the restaurants students selected. The most popular restaurant by far that students marked on our map was El Limon. This restaurant is just a couple of blocks away from campus, and one of the cheapest options in the area. Overall, stores and restaurants that were both closer to campus and relatively affordable had significantly more student patrons than stores that were more expensive or outside of the Ardmore area. These are two factors that seem to play a big role in how students go about engaging with food locations off campus.

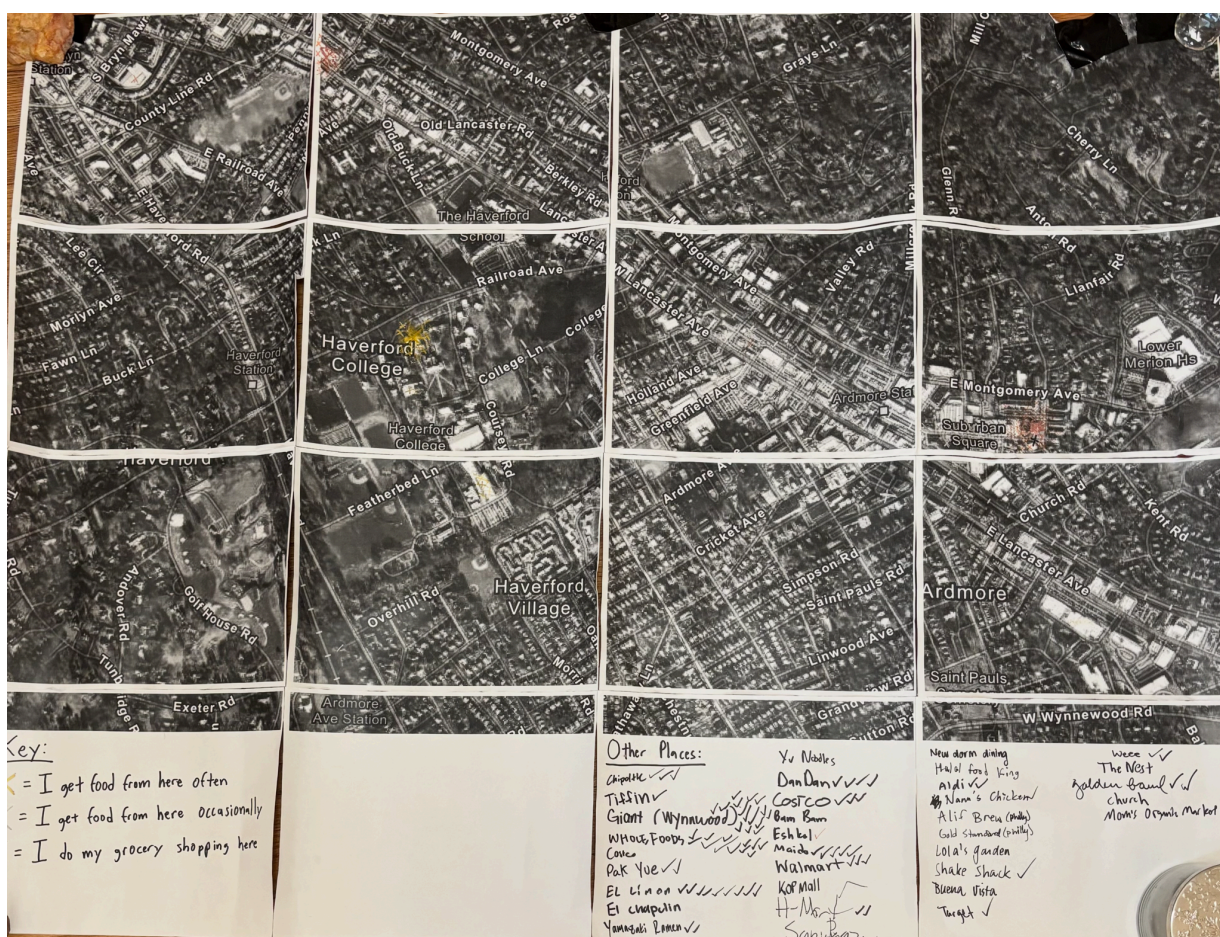


Figure 6: The Participatory Map with Student Responses

Discussions: Limitations and potential future directions

GIS Map

With only a little more than a month to put this project together, we did our best to present a full picture of the Bi-Co food system. That being said, our research was done in a short time frame and more detailed information could still be collected. For some of the places we have included in our study, we had to use the information we know from being students at these colleges and from doing inspectional research because the people who run certain spaces did not respond to our request to talk with them. We did reach out to students involved in certain places on campus if the direct managers did not answer, so using our own perspectives was a last resort.

There is information that we think is missing in our map, but we hope future students will conduct research in these areas. There are a couple of places where we did not talk to someone directly, including Wyndham at Bryn Mawr, the ECC Kitchen at Bryn Mawr, and the MCC Kitchen at Haverford. During the data collection process, we also came across some information that no one seemed to know the answer to. One example is the fact that several staff members at Bryn Mawr College did not know when UnCommon Grounds opened. If more time allowed, we would have spent more time at Special Collections, trying to go through historical documents to figure out when the cafe opened and how the idea of having a cafe on campus came about. There are also some big questions that we are left thinking about and did not answer in our map. One is what the relationship between locations on the two campuses looks like. We found out that there is not a lot of communication between locations, but we did not ask each location what places on

the other half of the Bi-Co they do communicate with. Future research could aim to address this question.

Budgets and financial decisions were another aspect that we did not specifically address during many of our interviews, so there are limitations to how much our map can inform the Bi-Co community about purchasing decisions both colleges make. More questions related to specifics about financial decisions should be added if interviews were to be conducted again. Figuring out how to access the budgets that dining services on both campuses have to submit each year would be an important first step. These budgets are supposed to be public, but we had difficulty finding them anywhere. With these food systems being so complex, there are many areas where future research could extend upon the information we collected.

Systems Map

Mapping the systematic relationships between all of the parts of the Bi-Co food system and uncovering patterns proved effective, but there are some serious limitations to this exercise as well that should be noted. First, we faced a time constraint that prevented us from digging deeper into each individual food location than we would've liked to. We had six weeks to put the entire project together which, when dealing with such a complex system, was too short to feel like we were able to accurately represent the system in our systems map. Additionally, we faced barriers to information because students like us are not privy to all of the information about what food is coming in and exact budgets. Some locations were very challenging to reach, and even when we did connect with them, they were not able to give us all of the relevant information. Also, due to Canva's slow processing speeds, the Haverford and Bryn Mawr systems had to be on separate tabs. In reality, they are quite interconnected particularly because Bryn Mawr

students can dine at Haverford, and Haverford students can dine at Bryn Mawr (and often do). Therefore, our model is missing those connections between the two systems.

However, as Bryn Mawr GIS professor Dirk Kinsey often says, “no model is perfectly accurate, but some are helpful.” This systems map is helpful for two main reasons: it allows us to identify gaps in the system, and it allows us to identify similar roles that some parts of the system share. We are able to look at the map, identify when parts are missing, and figure out what could be changed or added to fill in those gaps. For example, CER has a new composter down by the Haverford facilities building, turning unprocessed compost from HCA and the Phebe Anne Thorne preschool into processed compost. However, currently there isn’t a place where that compost is going. Seeing that on a systems map is quite easy because there isn’t an arrow leaving the composter that goes anywhere currently. Connecting that arrow to the Arboretum or the Haverfarm would help strengthen the system.

Additionally, we are able to look across campuses and identify parts of the system that are filling similar niches. Opening up a line of connection between them, could help both parts learn from each other. Maybe one group has learned how to deal with an issue that is very similar to one that another group is facing. For instance, through the systems map, we noticed that the HCA Garden Club and the BMC Community Garden are very similar groups. The HCA Garden Club is dealing with a lack of institutional knowledge and current momentum that the BMC Community Gardens are doing well with, while the BMC Community Garden is dealing with new infrastructure, and the HCA Garden Club has had theirs for a while now. Opening up communication between groups like this is a huge asset that this systems map gives us.

Participatory Map

If we were to do the participatory mapping project again, we would definitely change a few elements of our survey method to get clearer results. First, we would use a different system for tallying student votes that was easier to count. While the map was visually engaging and appealing, perhaps having a complete list of stores on the side including those within the confines of the map that students could then tally would be easier to analyze. We originally planned to use sticker dots; however, I wonder if in large quantities those would be as indistinguishable as “X”s.

Second, the times and locations we chose were correlated to when classes ended, so there would be lots of students moving around campus. That was very effective. However, for the apartment sampling we chose a Monday and Wednesday, which have similar student populations because classes that meet biweekly or triweekly are on the M/W/F schedule. If repeated, I think we should choose one Monday/Wednesday time and one Tuesday/Thursday time to get a greater sample of students. During our Wednesday sample, we noticed that many of the people walking past we had already talked to on Monday. The Dining Center sampling time was effective and did not seem to be skewed in the same way. Additionally, because we were only able to do participatory mapping on Haverford campus and this was a Bi-Co project, it would have been good to do a similar sampling section at the Bryn Mawr dining halls.

Lastly, we used color coding to distinguish the frequency with which people ate out, which I think would be interesting to implement for the grocery shopping tally. I got the sense that when people marked multiple locations for grocery shopping, there was usually one primary location where they shopped and then a secondary location where they supplemented their shopping. Also, people in the apartments have different degrees to which they are on the meal

plan. Some people eat every lunch at the dining hall because it is social and convenient, while some people are not on the meal plan at all. It would be interesting to color code the votes into “at least once a week”, “at least every other week”, and “at least once a month” to get a sense of how often people are going to various locations.

By addressing these limitations, future research can hopefully fill in the gaps that are present within our project.

Conclusion/Recommendations:

Food systems are always complicated, and while we don’t expect our food system on campus to be perfect, there are definitely ways to promote efficiency and better access that would make the system stronger. Whether it’s increasing the freshness of the ingredients or the affordability of the meal plans, improvements can always be made to on campus food systems, and our campuses are no different. Reflecting back on the goals of this project, we are proposing four broader considerations and recommendations for the future that span a wide range of feasibility and impact.

#1 - Improving communication between similar groups and locations

Starting at the ‘smallest’ or most easily influenced level of change, work can be done within and across student organizations. We have noticed that in particular, many student organizations are working parallelly towards similar goals, but are not connecting with each other in the process. For example, both Haverford and Bryn Mawr have student run gardens that are in the process of being established or revitalized: the HCA Garden and the BMC Community Garden. Information about growing practices such as what grows well in this climate, how to

navigate the administration to obtain funding, and strategies to leverage organizational networks are all necessary for the function of these locations. The HCA Garden in fact has a strong relationship with the Haverfarm to help pass on that information, but Bryn Mawr does not have an organizational equivalent that they can turn to. As uncovered through our systems map, we can also see that these organizations have the potential to share material resources as well. Building a network between these student gardens could increase the transfer of institutional knowledge and create a support system for when complications arise.

Additionally, through oral histories we have learned about the ways student clubs have supported each other's growth and knowledge in the past. The Haverford sustainability and justice-oriented living community at Haverford, Ehaus, was influential in helping Batten House at Bryn Mawr establish its operations in the 2000s. Since then, Batten House has grown into a very developed sustainability and justice-oriented living community of its own. Between some spotty student organization and hostility from the administration, Ehaus has not flourished in the same way since then. Now that Ehaus is being revamped and Batten has maintained institutional momentum, Batten has the opportunity to reciprocate that mentorship by sharing knowledge and resources. Despite the stated value that the administration has placed on student-led initiatives, they don't always back these initiatives up with financial and organizational support. While student groups should continue to ask for this support from the institution, it is important for them to simultaneously create their own networks of information and material sharing.

#2 - Increasing transparency and accuracy

In interactions between students and these institutions, actions can be taken to increase the transparency and accuracy of the information available to students about the food system

they are a part of. Our experiences gathering data for this project highlighted how difficult it is to get a clear picture of what is going on. First, almost no information is published on public websites, so we had to consult stakeholders themselves to access the information. However, knowing who to contact and what their contact information is is not straightforward, and we found ourselves emailing multiple people before we came into contact with the person who has the relevant information. This complex process also places a burden on institutional employees who already have to respond to many other emails, and they find themselves having the same conversations with student groups every couple of years as students cycle through campus and fail to pass on the information to new leaders.

There is also a question of whether the person who has been elected to represent a given location is representing it entirely accurately. For example, the Haverford Dining Center claims that it engages in pre-consumer composting; however, many student workers at the DC have not observed pre-consumer composting occurring. When a student group tried to measure the quantity of pre-consumer compost being created, they were unable to because the managers of the evening shift did not realize they were supposed to make the compost available to the student group. After multiple attempts and rounds of emailing, the student group redirected efforts. As previously stated, timely and accurate information is vital in determining what change can be made in the system. Increasing the amount of information available on websites, and transparency on who to contact, could significantly reduce the amount of emailing that needs to occur. Ensuring consistency in policies would also be necessary for increasing the accuracy of information.

#3 - Clarifying values

On an institutional level, values need to be clarified. There are certain elements of creating a sustainable food system that don't have a creative low cost solution, and may simply be expensive. For example, currently very little food in any dining hall in the Bi-Co is sourced locally because locally sourced food is significantly more costly than food from national providers. There are many systemic reasons locally sourced food is more expensive, one of which is that workers are usually paid a better and more livable wage on small scale farms than industrial operations. Locally sourced food cannot be as cheap as a provider like Sysco unless there is exploitation during the growing process at which point that negates the reason to 'buy local'. If Haverford or Bryn Mawr is truly invested in a sustainable future and food system, the budget will need to be redistributed. The common refrain that the administration tends to remind students of is that we are working within a constrained system, but we seem to forget that we have choices about how we direct our energy and resources. The money used for new athletics fields or a state of the art dorm could buy many, many years worth of locally sourced groceries. Money is finite, and there are many ways to spend that money on campus, but choosing not to use this money on food is a choice made by the colleges. Reconsidering these choices and diverting more money to improving the food system could help align us with the values we have expressed throughout the paper.

#4 - Adjusting policies

Also within the control of the institution are policies. In talking to student stakeholders, it appears that within recent years, institutional policies particularly surrounding food have become more rigid. This in turn has limited the action that students can take around food justice. For

example, because Haverford in recent years has tightened restrictions about students cooking and serving food, many of the events that Ehaus hosted pre-COVID are no longer possible. While many of these policies around student preparation of food are rooted in a concern for food safety, ultimately they increase food waste and dependency on buying cheap and unsustainably packaged snacks. In general, students across organizations feel discouraged to take action within the Bi-Co because policies are so rigid. Inordinate amounts of time can go into figuring out what school policies apply to a given initiative, and it discourages student groups to even try to host events or build community in this way. If these institutions could work to make policies less strict and more easily accessible, the sustainability and efficiency of the food system would benefit. Institutions should work with student groups to find mutually agreeable solutions, instead of simply saying rejecting an initiative or activity because it violates the policy.

What's next

In conclusion, the food system of The Bi-Co is complex, obviously. However, the complexity does not mean that action cannot or should not be taken. It is in fact the opposite because there are so many interconnected facets; changes within one piece of the system will ripple outwards. While it may not always feel like it, students do have agency within the system, particularly within the form of student organizations, and coalition building across organizations can be an effective way to build power. Throughout our project, we also have noticed how all actors within this system are involved in influencing the Bi-Co food system. Students, faculty, and staff need to work together to uphold a food system that supports everyone and the environment.

As a group of four seniors, we will not be around to enact or see the change that we know needs to happen. Our first and primary goal of this project was to consolidate information about these processes and progressions. Students can't contribute to change without having proper information about the issues. There is a lack of institutional memory at Bryn Mawr and Haverford due to constant student turnover, so we hope that future stakeholders can use the information we have collected to move more quickly to the changemaking stage. There is a lot of work to be done, we hope the next generation of Bryn Mawr and Haverford students will carry our work onwards!

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